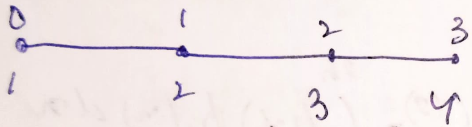
FDM

(1) $\dfrac{d^2u}{dx^2} - 2u = f(x)$, $0 < x < 1$, where $f(x) = 4x^2 - 2x - 4$

$u(0) = 0$ & $u(1) = -1$



for node 1

$$\frac{u_2 - 2u_1 + u_0}{h^2} - 2u_1 = f(x_1)$$

for node 2

$$\frac{u_3 - 2u_2 + u_1}{h^2} - 2u_2 = f(x_2)$$

$u_0 - (2+2h^2)u_1 + u_2 = h^2 f(x_1)$

$u_1 - (2+2h^2)u_2 + u_3 = h^2 f(x_2)$

$-(2+2h^2)u_1 + u_2 = h^2 f(x_1) - u_0$

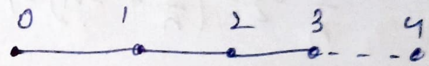
$u_1 - (2+2h^2)u_2 \cancel{+u_3} = h^2 f(x_2) - u_3$

$$\begin{bmatrix} -(2+2h^2) & 1 \\ 1 & -(2+2h^2) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} h^2 f(x_1) - u_0 \\ h^2 f(x_2) - u_3 \end{bmatrix}$$

$u_1 = 0.11111$

$u_2 = -0.2222$

for $u(0) = 0$ & $\frac{du}{dx}\Big|_1 = -3$



for node1

$$\frac{u_2 - 2u_1 + u_0}{h^2} - 2u_1 = f(x_1)$$

for node2

$$\frac{u_3 - 2u_2 + u_1}{h^2} - 2u_2 = f(x_2)$$

for node3

$$\frac{u_4 - u_2}{2\Delta x} = \frac{du}{dx} = -3$$

$u_4 = u_2 - 6h$

$$\frac{u_4 - 2u_3 + u_2}{h^2} - 2u_3 = f(x_3)$$

$u_4 - (2+2h^2)u_3 + u_2 = f(x_3)h^2$

$(u_2 - 6h) - (2+2h^2)u_3 + u_2 = h^2 f(x_3)$

$$\begin{bmatrix} -(2+2h^2) & 1 & \\ 1 & -(2+2h^2) & 1 \\ 0 & 2 & -(2+2h^2) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{bmatrix} h^2 f(x_1) - u_0 \\ h^2 f(x_2) \\ h^2 f(x_3) + 6h \end{bmatrix}$$

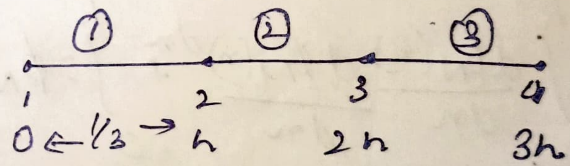
$u_1 = 0.11111$, $u_2 = -0.2222$, $u_3 = -1$

FEM

$$\frac{d^2u}{dx^2} - 2u = f(x), \quad 0 < x < 1 \qquad u(0) = 0, \quad u(1) = -1$$

$$f(x) = 4x^2 - 2x - 4$$

$$u = -2x^2 + x \qquad h = 1/3$$



$\cancel{\xi = \frac{x - x_j}{h}}$

$$\phi_1(x) = \frac{x_{e+1} - x}{h}, \quad \phi_2(x) = \frac{x - x_e}{h}$$

$\cancel{x = x_e + \xi h}$ $\quad x = x_e + sh$

$$\phi_1 = 1 - s, \quad \phi_2 = s$$

for element 1: $x_1 = 0, x_2 = h$

$$\phi_1(x) = \frac{h - x}{h}, \quad \phi_2(x) = \frac{x - 0}{h}$$

for element 2: $x_2 = h, x_3 = 2h$

$$\phi_1(x) = \frac{2h - x}{h}, \quad \phi_2(x) = \frac{x - h}{h}$$

for element 3: $x_3 = 2h, x_4 = 3h$

$$\phi_1(x) = \frac{3h - x}{h}, \quad \phi_2(x) = \frac{x - 2h}{h}$$

$$\phi_1'(x) = -1/h \,,\ \phi_2'(x) = 1/h$$

$$\int_0^h \phi_N^e \left(\frac{d^2u}{dx^2} - 2u\right) dx = \int_0^h f(x)\phi_N \, dx$$

$$= \int_0^h \frac{d^2u}{dx^2}\phi_N^e - \int_0^h 2u\phi_N^e \, dx = \int_0^h f(x)\phi_N^e \, dx$$

$$\phi_N^e \frac{du}{dx}\Big|_0^h - \int_0^h \frac{d\phi_N^e(x)}{dx}\frac{du^e(x)}{dx} dx - \int_0^h 2u\phi_N^e(x) dx = \int_0^h f(x)\phi_N^e(x) dx$$

$$\phi_N^e \frac{du}{dx}\Big|_0^h - \left[\int_0^h \frac{d\phi_N^e(x)}{dx}\frac{d\phi_M^e(x)}{dx} dx\right] u_M^e - \int_0^h 2u\phi_N^e(x) dx = \int_0^h f(x)\phi_N(x) dx$$

Rearranging the terms

$$\int_0^h \frac{d\phi_N(x)}{dx}\frac{d\phi_M(x)}{dx} dx + \int_0^h 2u\phi_N^e \, dx = \phi_N^e \frac{du}{dx}\Big|_0^h - \int_0^h f(x)\phi_N(x) dx$$

$$K_{NM} = \int_0^h \frac{d\phi_N(x)}{dx}\frac{d\phi_M(x)}{dx} dx + \int_0^h 2\phi_N(x) u \, dx$$

$$K_{11}^{(1)} = \int_0^h \left(\frac{d\phi_1(x)}{dx}\frac{d\phi_1(x)}{dx} + 2\phi_1\phi_1\right) dx$$

$$= \int_0^h \left(-\frac{1}{h}\right)\left(-\frac{1}{h}\right) dx = \frac{1}{h}$$

$$\int_0^h 2\phi_1\phi_1 \, dx = \frac{2h}{3}$$

$$\therefore K_{11}^{(1)} = \frac{1}{h} + \frac{2h}{3}$$

$$K_{12}^{(1)} = \int_0^h \frac{d\phi_1(x)}{dx}\frac{d\phi_2(x)}{dx}\,dx = \int_0^h (-1/h)(1/h)\,dx$$

$$= -1/h$$

$$2\int_0^h \phi_1\phi_2\,dx = \frac{2h}{6}$$

$$\therefore K_{12}^{(1)} = -\frac{1}{h} + \frac{2h}{6}$$

$$K_{21}^{(1)} = K_{12}^{(1)} = -\frac{1}{h} + \frac{2h}{6}$$

$$K_{22}^{(1)} = K_{11}^{(1)} = \frac{1}{h} + \frac{2h}{3}$$

$$\therefore K^{(1)} = \begin{bmatrix} 1/h + \frac{2h}{3} & -1/h + \frac{2h}{6} \\ -1/h + \frac{2h}{6} & \frac{1}{h} + \frac{2h}{3} \end{bmatrix}$$

$\therefore$ global K matrix is

$$K^{(1)} = \begin{bmatrix} \frac{1}{h} + \frac{2h}{3} & -\frac{1}{h} + \frac{2h}{6} & 0 & 0 \\ -\frac{1}{h} + \frac{2h}{6} & \left(\frac{1}{h} + \frac{2h}{3}\right) + \left(\frac{1}{h} + \frac{2h}{3}\right) & -\frac{1}{h} + \frac{2h}{6} & 0 \\ 0 & -\frac{1}{h} + \frac{2h}{6} & \left(\frac{1}{h} + \frac{2h}{3}\right) + \left(\frac{1}{h} + \frac{2h}{3}\right) & -1/h + \frac{2h}{6} \\ 0 & 0 & -1/h + \frac{2h}{6} & \frac{1}{h} + \frac{2h}{3} \end{bmatrix}$$

for case D-D case $u(0) = 0$, $u(1) = -1$

$$\begin{bmatrix} 0 \\ u_1 \\ u_2 \\ -1 \end{bmatrix} \quad f_1^{(1)} = \int_0^h f(x)\phi_1(x)\,dx \quad f_1^{(2)} = \int_h^{2h} f(x)\phi_1(x)\,dx$$

$$f_2^{(1)} = \int_0^h f(x)\phi_2(x)\,dx \quad f_2^{(2)} = \int_h^{2h} f(x)\phi_2(x)\,dx$$

$$f_1^{(3)} = \int_{2h}^{3h} f(x)\phi_1(x)\,dx$$

$$f_2^{(3)} = \int_{2h}^{3h} f(x)\phi_2(x)\,dx$$

$$\therefore F = \begin{bmatrix} F_1^{(1)} \\ F_2^{(1)} + F_1^{(2)} \\ F_2^{(2)} + F_1^{(3)} \\ F_2^{(3)} \end{bmatrix} = \begin{bmatrix} 0.6913580 \\ 1.3827160 \\ 1.160493 \\ 0.43209 \end{bmatrix}$$

$\therefore Ku = F$

$\therefore$ ~~$u_1 = 0.1108$~~ $u_1 = 0.1180556$

$u_2 = -0.2152778$

for

$u(0) = 0$ and $\left.\frac{dy}{dx}\right|_1 = -3$

using $Ku = F + G$

$$G = \begin{bmatrix} G_1^{(1)} \\ G_2^{(1)} + G_1^{(2)} \\ G_2^{(2)} + G_1^{(3)} \\ G_2^{(3)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3 \end{bmatrix}$$

$\therefore$ $u_1 = 0.12317452$

$u_2 = -0.20392548$

$u_3 = -0.97976461$

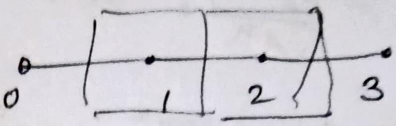
FVM

$$\frac{d^2u}{dx^2} - 2u = f(x), \quad 0 < x < 1$$

$u(0) = 0$

$u(1) = -1$

$$f(x) = 4x^2 - 2x - 4$$



$x_0 = 0,\ x_1 = 1/3,\ x_2 = 2/3,\ x_4 = 1$

$$\int_{i-1/2}^{i+1/2} u''\,dx - 2\int_{i-1/2}^{i+1/2} u\,dx = \int_{i-1/2}^{i+1/2} f(x)\,dx$$

$$\left.\frac{du}{dx}\right|_{i-1/2}^{i+1/2} - 2u\Big|_{i-1/2}^{i+1/2} = f(x_i)\,\Delta x\Big|_{i-1/2}^{i+1/2}$$

$$\left.\frac{du}{dx}\right|_{i+1/2} = \frac{u_{i+1} - u_i}{x_{i+1} - x_i} = \frac{u_{i+1} - u_i}{h}$$

$$\left.\frac{du}{dx}\right|_{i-1/2} = \frac{u_i - u_{i-1}}{x_i - x_{i-1}} = \frac{u_i - u_{i-1}}{h}$$

for node 1

$$\left(\frac{u_2 - u_1}{h} - \frac{u_1 - u_0}{h}\right) - 2u_1 h = f(x_1) \times h$$

$$\frac{u_2 - 2u_1 + u_0}{h} - 2u_1 h = h f(x_1)$$

$$\frac{u_2}{h} - \left(\frac{2}{h} + 2h\right)u_1 + \frac{u_0}{h} = h f(x_1)$$

$$\frac{u_2}{h} - \left(\frac{2}{h} + 2h\right)u_1 = h f(x_1) - \frac{u_0}{h}$$

for node 2

$$\left(\frac{u_3 - u_2}{h}\right) - \left(\frac{u_2 - u_1}{h}\right) - 2u_2 h = h f(x_2)$$

$$\frac{u_3 - 2u_2 + u_1}{h} - 2u_2 h = h f(x_2)$$

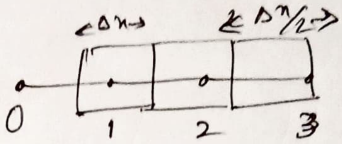
$$\frac{U_3}{h} - \left(\frac{2}{h} + 2h\right) U_2 + \frac{U_1}{h} = h f(x_2)$$

$$-\left(\frac{2}{h} + 2h\right) U_2 + \frac{U_1}{h} = h f(x_2) - \frac{U_3}{h}$$

$\therefore$ solving for u

$$\begin{bmatrix} -(2/h + 2h) & (1/h) \\ (1/h) & -(2/h + 2h) \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \end{Bmatrix} = \begin{bmatrix} h f(x_1) - (U_0/h) \\ h f(x_2) - (U_3/h) \end{bmatrix}$$

$\therefore U_1 = 0.11111$
$\quad\; u_2 = -0.2222$



case 2
for $u(0) = 0$ & $\left.\frac{du}{dx}\right|_{x=1} = -3$

at node 3

$$\left.\frac{du}{dx}\right|_{x=1} - \left.\frac{du}{dx}\right|_{i-1/2} - 2u_i\left(\frac{h}{2}\right) = f(x_3)\left(\frac{h}{2}\right)$$

$$-3 - \left(\frac{U_3 - U_2}{h}\right) - 2U_3\left(\frac{h}{2}\right) = f(x_3)\left(\frac{h}{2}\right)$$

$$-(1+h^2)U_3 + U_2 = f(x_3)\frac{h^2}{2} + 3h$$

$$\rightarrow \boxed{-2(1+h^2)U_3 + 2U_2 = f(x_3)h^2 + 6h}$$

at node 2

$$\frac{U_3 - 2U_2 + U_1}{h} - 2U_2 h = h f(x_2)$$

$$\therefore U_3 - (2 + 2h^2)U_2 + U_1 = h^2 f(x_2)$$

at node 1

$$U_2 - (2 + 2h^2)U_1 + U_0 = h^2 f(x_1)$$

solving for u,

$$\begin{bmatrix} -(2+2h^2) & 1 & 0 \\ 1 & -(2+2h^2) & 1 \\ 0 & 2 & -2(1+h^2) \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} = \begin{bmatrix} h^2 f(x_1) - U_0 \\ h^2 f(x_2) \\ h^2 f(x_3) + 6h \end{bmatrix}$$

$\therefore u_1 = 0.1111, \; u_2 = -0.2222, \; u_3 = -\frac{1}{2}$